TU DELFT

Development Kit

Applied Experimental Methods

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This document is intended to help you set up the haptic hardware kit used in the Applied Experimental Methods course at TU Delft. The kit is based on the BLACK PILL STM32F401X microcontroller and can be programmed using the Arduino IDE. In this document, you will find example programs to assist you in creating your own experimental setup. For any questions not handled in this document, please reach out to the course assistants.

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1 Introduction

This document provides a practical guide for setting up and using the haptic development kit employed in the *Applied Experimental Methods* course at TU Delft. The kit is centered around the Black Pill STM32F411 microcontroller, which serves as the main controller for reading sensors, interfacing with peripheral boards, and driving different actuators.

The purpose of this manual is to help students become familiar with the hardware included in the kit and to provide a starting point for building their own experimental setups. The system includes several sensing and actuation components, as well as interface and conditioning boards that allow the microcontroller to communicate with mixed-signal peripherals. Through this combination of modules, the kit supports the development of experiments involving force, temperature, inertial sensing, vibrotactile feedback, and thermal actuation.

Figure 1.1 shows the complete haptic development kit used throughout this document.

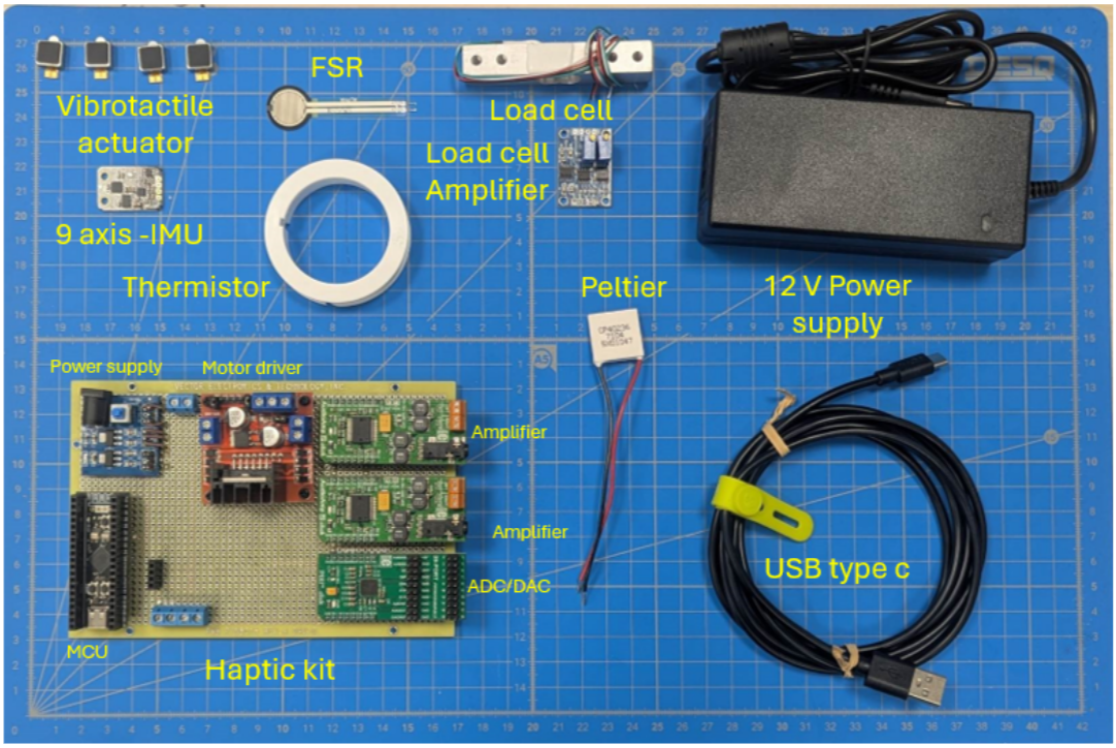

Figure 1.1: Overview of the haptic development kit used in the Applied Experimental Methods course.

2 Black Pill STM32F411

The Black Pill STM32F411 is the main microcontroller board used in this setup. It is based on the STM32F411CEU6 microcontroller, which features an Arm Cortex-M4 core running at up to 100 MHz, 512 KB of Flash memory, and 128 KB of SRAM. The board provides digital GPIO pins, analog inputs, PWM-capable pins, and communication interfaces such as I^2C, SPI, and UART, which makes it suitable for reading sensors and controlling external modules.

In this setup, the Black Pill acts as the central controller and communicates with the Pixi Click board, which is then used as the configurable analog and digital interface for several of the connected devices. The Black Pill can be powered through USB-C or through its external power pins, while its GPIO pins are used for communication and control. The exact pins used depend on the peripherals connected in the final system, but typical examples include I^2C lines for communication, SPI lines for the Pixi Click interface, and digital outputs for actuator control.

The board operates at 3.3 V logic level, so connected peripherals must be compatible with this voltage level or must be interfaced appropriately. Depending on the board version, power can be supplied through USB, 5 V pins, or regulated 3.3 V pins.

For more information check the product pages at these links: `https://www.st.com/resource/en/datasheet/stm32f411ce.pdf` and `https://stm32-base.org/boards/STM32F411CEU6-WeAct-Black-Pill-V2.0.html`

2.1 Black Pill STM32F401x in Arduino IDE

In this section, you will learn how to set up the Black Pill STM32F411 board in the Arduino IDE, install the required board package, choose the correct board settings, and upload your first test program. Although this manual starts with Arduino IDE, the Black Pill can also be used with other IDEs and programming approaches if needed.

Before starting, make sure you have:

- A Black Pill STM32F411 board
- A USB to USB-C cable
- Arduino IDE 2 installed on your computer

The procedure below describes how to configure the Arduino IDE and upload code to the Black Pill STM32F411 board using DFU mode.

1. Open the Arduino IDE.
2. Open the Preferences menu: Go to `File > Preferences` or `Arduino IDE > Settings`, depending on your operating system.

3. Add the STM32 Board Manager URL: In the `Additional Boards Manager URLs` field, add the following link: `https://github.com/stm32duino/BoardManagerFiles/raw/main/package_stmicroelectronics_index.json`

4. Install the STM32 board package: Go to `Tools > Board > Boards Manager`, search for `STM32 MCU based boards`, and install the latest version published by STMicroelectronics.

5. Install STM32CubeProgrammer: Download and install STM32CubeProgrammer from `https://www.st.com/en/development-tools/stm32cubeprog.html`. This software is required for uploading code to the STM32 board when using `STM32CubeProgrammer (DFU)` as the upload method.

6. Select the correct board settings: After installation, go to `Tools` and configure the board as follows:

   - **Board:** Generic STM32F4 series
   - **Board part number:** BlackPill F411CE
   - **Upload method:** STM32CubeProgrammer (DFU)
   - **USB support:** CDC (generic 'Serial' supersede U(S)ART)
   - **USB speed:** Low/Full Speed

7. Connect the board to the computer using a USB-C to USB cable.

8. Put the board into DFU mode: To upload code through DFU, the board must first be placed into DFU mode. The recommended method is to disconnect the board from the computer, press and hold the `BOOT` button, reconnect the board through USB while keeping the button pressed, and then release the `BOOT` button once the board is connected.

   An alternative method is to press and hold the `BOOT` button, briefly press and release the `RESET` button, and then release the `BOOT` button. However, this method is less reliable and may result in inconsistent device detection by the computer.

9. Open the `Tools` menu in the Arduino IDE and check whether the board is detected correctly. If a port associated with the board appears under `Port`, select it before proceeding. If no port appears at this stage, continue with the upload process; after the code is uploaded successfully, the board will typically appear as a serial device.

10. Upload the code to the board.

2.2 Testing the Board with a Simple Program

The first program uploaded to the board should be as simple as possible. Its purpose is to confirm that the board is correctly detected, the code compiles without errors, the upload process works properly, and the microcontroller is running as expected.

One of the easiest programs for this initial verification is a basic LED blink example, shown below:

```
#define LED_PIN PC13

void setup() {
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(115200);
  delay(1000);

  Serial.println("STM32 Black Pill is starting...");
}

void loop() {
  digitalWrite(LED_PIN, LOW);   // LED on
  Serial.println("LED ON");
  delay(500);

  digitalWrite(LED_PIN, HIGH);  // LED off
  Serial.println("LED OFF");
  delay(500);
}
```

This is one of the simplest and most commonly used test programs for checking whether the board is working correctly. However, other basic examples can also be used for this purpose, such as simple serial communication tests or programs that read a digital input. Students are encouraged to explore additional beginner-level examples in Arduino IDE once the board has been successfully configured.

1. Upload the example program shown above, or another simple test program of your choice.
2. For this example, the onboard LED should blink continuously. On Black Pill boards, the onboard LED is typically connected to `PC13`. However, when supported by the selected board definition, it is preferable to use `LED_BUILTIN`.
3. Before each upload, the board must be placed into DFU mode. To do this, press and hold the `BOOT` button, then connect the board to the computer via USB while keeping the button pressed. Once connected, you can release the `BOOT` button.
4. If the selected program requires external libraries, install them first in Arduino IDE by going to `Sketch > Include Library > Manage Libraries`. Search for the required

library, install it, and include it in the code using the corresponding `#include` statement.

5. After the upload is complete, verify that the program behaves as expected. For the LED blink example, confirm that the onboard LED starts blinking. If the program also sends messages through the serial port, open the Serial Monitor and check that they are displayed correctly.

6. If the expected behavior is observed, the board has been successfully configured and is ready for further development.

3 Control Peripherals

3.1 PIXI Click Interface Board

The PIXI Click board is used in this kit as an interface between the STM32 Black Pill microcontroller and mixed-signal peripherals. It is based on the MAX11300 mixed-signal data converter, which provides configurable analog and digital input/output functions. This makes it suitable for acquiring signals from external circuits and for supporting communication between the microcontroller and additional sensing or conditioning stages.

In this setup, the PIXI Click board communicates with the STM32 Black Pill through the SPI interface. The SPI lines are used for data exchange between the microcontroller and the MAX11300, while additional control pins such as chip select, reset, interrupt, and conversion trigger are used for device control and synchronization. Through this interface, the Black Pill can configure the ports of the PIXI Click board and read or write signals depending on the connected peripheral.

Before integrating the PIXI Click board into a complete measurement system, it is recommended to first establish and verify its communication with the STM32 Black Pill, in order to ensure that the SPI interface and the control pins are working correctly.

For more information check the product page at this link: `https://www.mikroe.com/pixi-click`

3.1.1 Connection

The PIXI Click board is connected to the STM32 Black Pill through the SPI interface, together with the required power and control signals.

The required signal groups are:

- **SPI communication lines:** MOSI, MISO, SCLK, and Chip Select
- **Control lines:** CNVT and INT

- **Power lines:** 3.3V, 5V, and GND

Important: The PIXI Click board communicates through SPI, not through I^2C. Therefore, the `SDA` and `SCL` pins of the STM32 Black Pill are not used for this interface.

3.1.2 Libraries and Software Support

The PIXI Click board is based on the MAX11300 mixed-signal converter and communicates through SPI. This board requires software configuration of its internal ports before it can be used, since each port can be programmed individually for functions such as analog input, analog output, digital input, digital output, or analog switching.

The provided Arduino library simplifies this process by abstracting low-level communication details and offering easy-to-use functions to configure pins and read/write data.

Download the ZIP file of the library from the following link:... Then, in the Arduino IDE, go to `Sketch > Include Library > Add .ZIP Library` and select the downloaded file to install it.

To verify that the communication between the Pixi Click and the STM32 microcontroller is working correctly, follow the code below to configure one port as an analog input and read its value. If the pin is connected to GND, the measured value should be close to 0, while if it is connected to Vcc, the reading should be close to the maximum value. If the pin is left unconnected, the readings will fluctuate due to electrical noise.

```

#include "MAX11300.h"

#define CNVT_PIN PB0
#define CS_PIN   PA4

MAX11300 pixi(&SPI, CNVT_PIN, CS_PIN);

const uint8_t IN_PIN = 6;

void setup() {
  Serial.begin(115200);
  delay(1000);

  pixi.begin();

  pixi.setPinMode(IN_PIN, analogIn);
  pixi.setPinADCrange(IN_PIN, ADCZeroTo2_5);
  pixi.setPinADCref(IN_PIN, ADCInternal);
  pixi.setADCmode(ContinuousSweep);

  Serial.println("Setup done");
```

```
}

void loop() {
  uint16_t val = pixi.readAnalogPin(IN_PIN);
  Serial.print("Read Analog: ");
  Serial.println(val);
}
```

NOTE: The MAX11300 integrates a single ADC that is shared across all configured analog input channels through time multiplexing. As a result, the specified maximum conversion rate (up to 400 kSPS) represents the total sampling rate, which is distributed among all active channels. Therefore, the effective sampling rate per channel decreases as the number of active channels increases.

In addition, the ADC supports sample averaging, where multiple consecutive measurements are combined to produce a single output value.

3.2 Audio Amplifier Click Board

The AudioAmp 2 Click is an audio amplifier expansion board from MikroElektronika based on the TDA7491, a dual BTL class-D audio amplifier integrated circuit. It is intended for audio amplification applications and can deliver up to 20 W to an 8 Ω load under appropriate operating conditions. The board includes a 3.5 mm stereo line input, left and right speaker output terminals, selectable gain settings, mute and standby control, and a diagnostic output for fault monitoring.

For more information check the product pages at these links: `https://www.mikroe.com/audioamp-2-click` and
`https://download.mikroe.com/documents/datasheets/TDA7491HV.pdf`

3.2.1 Connections

The AudioAmp 2 Click can be controlled through the Pixi Click board, which is based on the MAX11300 programmable mixed-signal I/O integrated circuit. In this configuration, the Pixi Click is used to provide the digital control lines for the amplifier, while the audio signal itself must still be provided separately through the 3.5 mm line input. This is because the AudioAmp 2 Click does not use I^2C or SPI for normal operation, but instead relies on GPIO-based control pins such as gain selection, mute, and standby.

- **Black Pill 3.3 V or 5 V → AudioAmp 2 Click power supply**
- **Black Pill GND → AudioAmp 2 Click ground**
- **Pixi Click MAX11300 GPO → AudioAmp 2 Click GN0**

- **Pixi Click MAX11300 GPO → AudioAmp 2 Click GN1**
- **Pixi Click MAX11300 GPO → AudioAmp 2 Click STB**
- **Pixi Click MAX11300 GPO → AudioAmp 2 Click MUTE**
- **AudioAmp 2 Click DIA → Pixi Click MAX11300 input or directly to a Black Pill input**
- **Audio source / waveform source → 3.5 mm line input**
- **Load / actuator / speaker → LEFT or RIGHT output terminal**

In this setup, the Pixi Click is used only for digital control and optional status monitoring. The audio waveform that is to be amplified must be generated separately and fed into the line input of the AudioAmp 2 Click. If the board is used to drive a voice coil haptic actuator, the actuator should be connected across one amplifier output channel, while the control pins can still be managed through the Pixi Click.

3.2.2 Practical notes

1. The AudioAmp 2 Click is designed as an audio amplifier, so it is most suitable for loads that require an AC excitation signal.
2. The output is a BTL output stage, so the load should be connected directly across the output terminals and not referenced to ground.
3. The Pixi Click can be used for control signals such as gain, mute, and standby, but it does not replace the need for a separate audio waveform input.
4. The gain configuration should be selected carefully to avoid excessive output amplitude and possible overdrive of the connected actuator or speaker.
5. If this board is used with a haptic actuator, the actuator electrical limits must be checked carefully before operation.

3.3 L298N Motor Driver Board

The L298N Motor Driver Board is a driver module based on the ST L298 dual full-bridge driver integrated circuit. It can be used to control two DC motors or one stepper motor, and the product page specifies a current capability of 2 A per channel. The underlying L298 device is a dual full-bridge driver intended for inductive loads such as relays, solenoids, DC motors, and stepper motors.

For more information check the product page at these links: `https://www.benselectronics.nl/l298n-motor-driver-board.html`

3.3.1 Connections

The L298N Motor Driver Board can be controlled through the Pixi Click board, which is based on the MAX11300 programmable mixed-signal I/O integrated circuit. In this configuration, the Pixi Click provides the digital control signals for the L298N, while the external power supply provides the load current. Since the L298N is a full-bridge driver, one channel typically uses two logic inputs to define the output polarity and one enable input to activate the channel or apply PWM control.

- **External supply positive → L298N motor supply input**
- **External supply ground → L298N power ground**
- **Pixi Click MAX11300 GPO → L298N IN1**
- **Pixi Click MAX11300 GPO → L298N IN2**
- **Pixi Click MAX11300 GPO → L298N IN3**
- **Pixi Click MAX11300 GPO → L298N IN4**
- **Pixi Click MAX11300 GPO → L298N ENA** *(optional, if enable is controlled externally)*
- **Load terminal 1 → L298N OUT1**
- **Load terminal 2 → L298N OUT2**
- **Load terminal 3 → L298N OUT3**
- **Load terminal 4 → L298N OUT4**

The four control signals connected to inputs in the Motor Driver define the polarity applied to the load. If the **ENA** input is also driven from the Pixi Click, the channel can be enabled or disabled by software. If the board is used with the onboard enable jumper left in place, only the inputs need to be controlled. This makes the Pixi Click suitable as a flexible logic interface between the Black Pill and the L298N board.

3.3.2 Practical notes

1. The product page states that the module can be used for **2 x DC motors or 1 x stepper motor** and specifies **2 A per channel**.
2. The L298 is a **dual full-bridge driver**, so it is suitable for bidirectional control of DC loads.
3. The load current must be supplied from an external power source; the Black Pill and the Pixi Click provide only logic-level control signals. This is an engineering consequence of the L298 being a separate power driver stage. :

4. The L298 introduces a noticeable voltage drop across the output stage, so the voltage seen by the load will be lower than the applied supply voltage. This follows from the bipolar full-bridge architecture described in the L298 datasheet.

4 Sensing Peripherals

4.1 10-DOF IMU Module

The IMU module used in this kit is the **DFRobot Fermion 10 DOF IMU Sensor V2 (SEN0140)**. Unlike single-chip IMUs, this module combines four different sensors on the same board and communicates through the I^2C interface. The integrated sensors are:

- **ADXL345**: 3-axis accelerometer
- **ITG3200 / ITG3205**: 3-axis gyroscope
- **VCM5883L**: 3-axis magnetometer
- **BMP280**: barometric pressure and temperature sensor

Because the module contains several independent devices, the recommended workflow is to test each sensor separately before combining all measurements in one program.

For more information check the product pages at these links:
`https://www.dfrobot.com/product-818.html`

4.1.1 Connections

Connect the IMU module to the STM32 Black Pill board using the I^2C interface:

- **VCC → 3.3V**
- **GND → GND**
- **SDA → I^2C SDA pin**
- **SCL → I^2C SCL pin**

Important: Even though the module includes an onboard regulator and accepts a wider power input range, the STM32 Black Pill operates with 3.3V logic, so using 3.3V is the safest option for I^2C communication.

4.1.2 Libraries

For this module, use the libraries referenced in the DFRobot documentation.

- **For orientation:**
  - DFRobot_ADXL345
  - DFRobot_ITG3200
  - DFRobot_QMC5883
- **For barometer and temperature measurements:**
  - DFRobot_BMP280

These libraries should be downloaded and installed in the Arduino IDE before uploading the example programs.

4.1.3 Examples

Barometer and temperature test

The following example verifies communication with the BMP280 sensor integrated in the IMU module. It prints temperature, pressure, and estimated altitude values to the Serial Monitor.

```
#include <Wire.h>
#include "DFRobot_BMP280.h"

typedef DFRobot_BMP280_IIC BMP;
BMP bmp(&Wire, BMP::eSdoLow);

#define SEA_LEVEL_PRESSURE 1013.25f

void setup() {
  Serial.begin(115200);
  delay(1000);
  Serial.println("BMP280 demo on STM32 Black Pill");

  Wire.begin(PB7, PB6);   // SDA, SCL

  while (bmp.begin() != BMP::eStatusOK) {
    Serial.println("BMP280 init failed, check wiring.");
    delay(2000);
  }

  Serial.println("BMP280 initialized correctly.");
}

void loop() {
  float temperature = bmp.getTemperature();
```

```
  uint32_t pressure = bmp.getPressure();
  float altitude = bmp.calAltitude(SEA_LEVEL_PRESSURE, pressure);

  Serial.print("Temperature = ");
  Serial.print(temperature);
  Serial.println(" *C");

  Serial.print("Pressure = ");
  Serial.print(pressure);
  Serial.println(" Pa");

  Serial.print("Altitude = ");
  Serial.print(altitude);
  Serial.println(" m");

  Serial.println();
  delay(2000);
}
}
```

After uploading the code, open the Serial Monitor to verify that the temperature, pressure, and altitude values are printed correctly.

Combined IMU example

This example reads data from the different sensors integrated in the IMU module and displays the results in the Serial Monitor. The accelerometer values are used to estimate pitch and roll, the gyroscope provides angular velocity along the three axes, the magnetometer is used to estimate the heading, and the BMP280 provides temperature, pressure, and altitude measurements.

In this way, different information about the motion and orientation of the system can be observed, as well as the surrounding environmental conditions.

```
#include <Wire.h>
#include <math.h>

#include "DFRobot_ADXL345.h"
#include "DFRobot_ITG3200.h"
#include "DFRobot_BMP280.h"
#include "DFRobot_QMC5883.h"

DFRobot_ADXL345 accel;
DFRobot_ITG3200 gyro;
DFRobot_QMC5883 compass;

```

```
typedef DFRobot_BMP280_IIC BMP;
BMP bmp(&Wire, BMP::eSdoLow);

#define SEA_LEVEL_PRESSURE 1013.25f

void setup() {
  Serial.begin(115200);
  delay(1000);

  Wire.begin(PB7, PB6);

  if (!accel.begin()) {
    Serial.println("ADXL345 init failed");
    while (1);
  }

  if (!gyro.begin()) {
    Serial.println("ITG3200 init failed");
    while (1);
  }

  if (!compass.begin()) {
    Serial.println("Magnetometer init failed");
    while (1);
  }

  while (bmp.begin() != BMP::eStatusOK) {
    Serial.println("BMP280 init failed");
    delay(2000);
  }

  gyro.zeroCalibrate(128, 5);

  Serial.println("All sensors initialized correctly.");
}

void loop() {
  int xyz[3];
  accel.readAccel(xyz);

  float ax = xyz[0];
  float ay = xyz[1];
  float az = xyz[2];

  float roll  = atan2(ay, az) * 180.0 / PI;
  float pitch = atan2(-ax, sqrt(ay * ay + az * az)) * 180.0 / PI;

  float gx, gy, gz;
  gyro.readGyro(&gx, &gy, &gz);
```

```
  Vector mag = compass.readNormalize();

  float heading = atan2(mag.YAxis, mag.XAxis) * 180.0 / PI;
  if (heading < 0) heading += 360.0;

  float temperature = bmp.getTemperature();
  uint32_t pressure = bmp.getPressure();
  float altitude = bmp.calAltitude(SEA_LEVEL_PRESSURE, pressure);

  Serial.println("===== SENSOR DATA =====");

  Serial.print("Accel X: "); Serial.print(ax);
  Serial.print("  Y: "); Serial.print(ay);
  Serial.print("  Z: "); Serial.println(az);

  Serial.print("Roll: "); Serial.print(roll);
  Serial.print(" deg   Pitch: "); Serial.print(pitch);
  Serial.println(" deg");

  Serial.print("Gyro X: "); Serial.print(gx);
  Serial.print("  Y: "); Serial.print(gy);
  Serial.print("  Z: "); Serial.println(gz);

  Serial.print("Mag X: "); Serial.print(mag.XAxis);
  Serial.print("  Y: "); Serial.print(mag.YAxis);
  Serial.print("  Z: "); Serial.println(mag.ZAxis);

  Serial.print("Heading: "); Serial.print(heading);
  Serial.println(" deg");

  Serial.print("Temperature: "); Serial.print(temperature);
  Serial.println(" *C");

  Serial.print("Pressure: "); Serial.print(pressure);
  Serial.println(" Pa");

  Serial.print("Altitude: "); Serial.print(altitude);
  Serial.println(" m");

  Serial.println();
  delay(500);
}
```

Yaw, pitch, and roll estimation

This example estimates yaw, pitch, and roll from the IMU module and displays the values in the Serial Monitor. By moving the sensor in different directions, it can be observe how the estimated orientation changes in real time.

Important: For this example, the `FreeSixIMU` library must be installed manually, since it is not included in the standard Arduino Library Manager. The library can be downloaded from the following repository: `https://github.com/DFRobot/FreeSixIMU`.

```

#include <FreeSixIMU.h>
#include <FIMU_ADXL345.h>
#include <FIMU_ITG3200.h>

#include <Wire.h>

float angles[3]; // yaw pitch roll

// Set the FreeSixIMU object
FreeSixIMU sixDOF = FreeSixIMU();

void setup() {
  Serial.begin(9600);
  Wire.begin();

  delay(5);
  sixDOF.init(); //begin the IMU
  delay(5);
}

void loop() {

  sixDOF.getEuler(angles);

  Serial.print(angles[0]);
  Serial.print(" | ");
  Serial.print(angles[1]);
  Serial.print(" | ");
  Serial.println(angles[2]);

  delay(100);
}
```

After uploading the code, open the Serial Monitor to observe the yaw, pitch, and roll values. As the IMU module is rotated, the displayed orientation values should change accordingly.

4.2 Load Cell with AD620 Signal Amplifier Module

The load cell is a force sensor that converts mechanical deformation into a very small differential electrical signal. Since this output signal is too small to be read directly with sufficient resolution by the Pixi Click board, it is used together with the AD620 instrumentation amplifier module. The AD620 amplifies the differential voltage generated by the load cell and provides a single-ended analog output that can be measured by the Pixi Click board.

In this setup, the load cell is excited using an external 5 V supply. The red wire is connected to the positive excitation voltage, while the black wire is connected to ground. The differential output wires of the load cell, green and white, are connected to the `S+` and `S-` inputs of the AD620 module, respectively. The AD620 module is powered from an external supply, and its output `Vout` is connected to the analog input pin of the Pixi Click board. The grounds of the amplifier, the load cell excitation source, and the Pixi Click board must all be common.

The amplification level can be adjusted using the onboard gain potentiometer of the AD620 module, while the offset can also be tuned if needed. This is useful to adapt the sensor output to the analog input range of the Pixi Click board and to improve the measurement resolution. Since the output of the amplifier depends on the selected gain and supply voltage, the output should be adjusted carefully to remain within the acceptable input range of the Pixi Click board.

For more information check the product pages at these links `https://www.tinytronics.nl/en/communication-and-signals/io-converters/signal-amplifier-module-ad620` and `https://www.otronic.nl/nl/weegcel-yzc-131-sensor-tot-1-kg.html`

4.3 FSR

An FSR is a two-terminal sensor whose resistance decreases as the applied force increases. To obtain a measurable signal, the FSR is used in a voltage divider configuration with a fixed resistor, which converts the resistance variation into a voltage that can be read by an analog input. In this setup, a 10 kΩ pull-up resistor is externally soldered, so no additional resistor is required.

The FSR is connected between the analog input pin of the Pixi Click board and GND, while the pull-up resistor connects the same input node to Vcc (3.3V). As the applied force increases, the resistance of the FSR decreases, resulting in a change in the voltage at the input, which can then be measured by the Pixi Click board.

For more information check the product pages at these links: `https://www.kiwi-electronics.com/image/catalog/pdf/FSR402.pdf`

4.4 Thermistor

The thermistor is a two-terminal temperature sensor whose resistance varies with temperature. In this case, a Negative Temperature Coefficient (NTC) thermistor is used, meaning that its resistance decreases as temperature increases.

Similarly to the FSR, the thermistor is usually used in a voltage divider configuration. In this setup, a 22 kΩ pull-up resistor is externally soldered, forming a divider that converts resistance changes into a measurable voltage.

The thermistor is connected between the analog input pin of the Pixi Click board and GND, while the pull-up resistor connects the same node to Vcc (3.3V). As temperature increases, the resistance of the thermistor decreases, resulting in a change in the voltage at the input, which can then be measured by the Pixi Click board.

The relationship between resistance and temperature is nonlinear, as shown in the datasheet, and typically requires calibration or a lookup table for accurate temperature estimation.

For more information check the product pages at these links `https://www.semitec-global.com/uploads/2022/01/P8-F-micro-Thermistor.pdf`

5 Actuators

5.1 Voice Coil Haptic Actuator

The Tactile Labs HapCoil-One, also referred to by the manufacturer as the Haptuator Mark II-D, is a voice coil haptic actuator intended for broadband vibrotactile feedback. Unlike a linear resonant actuator, which is typically driven near a single resonant frequency, this actuator can operate over a wider frequency range and can be driven in a manner similar to a loudspeaker. According to the manufacturer, it provides large rated bandwidth, high acceleration output, and compatibility with most audio amplifiers.

For more information check the product pages at these links `https://tactilelabs.com/product/hapcoil-one/`

5.1.1 Connections

MISSING

5.1.2 Practical notes

1. The HapCoil-One should not be driven directly from a GPIO pin, since it requires an external amplifier stage.

2. This actuator is better suited for broadband vibrotactile rendering than a narrowband LRA, since the manufacturer describes it as having large rated bandwidth.
3. The final vibration response will depend strongly on the way the actuator is mounted in the mechanical structure, so the integration should be designed carefully. This is a general engineering consideration for voice coil haptic actuators.

5.2 Wide Band Linear Resonant Actuator

The Vybronics VLV101040A is a linear resonant actuator (LRA) intended for haptic feedback applications. Unlike an eccentric rotating mass motor, an LRA generates vibration by oscillating an internal mass near its resonant frequency, which enables faster response and more controlled haptic output. This actuator has dimensions of 10 × 10 × 4 mm, a Z-axis vibration direction, a rated voltage of 2.5 V_{RMS}, an operating voltage range of 0.1 to 2.5 V_{RMS}, a resonant frequency of 170 Hz, a rated current of 350 mA maximum, a typical current of 317 mA, and a vibration force of 2.75 G under rated conditions.

For more information check the product pages at these links: `https://www.vybronics.com/wp-content/uploads/datasheet-files/Vybronics-VLV101040A-datasheet.pdf`

5.2.1 Connections

MISSING

5.2.2 Practical notes

1. The VLV101040A should not be driven directly from a GPIO pin, since it requires an AC excitation signal rather than a simple DC drive.
2. Best performance is achieved when the actuator is driven close to its resonant frequency of 170 Hz.
3. Because this actuator has a wide bandwidth, haptic drivers using auto-resonance detection may require manual configuration.
4. The mounting condition affects the vibration response, so the actuator should be fixed carefully in the final mechanical structure. This part is designed for pressure contacts such as pogo pins or spring contacts.

5.3 Coin-type Linear Resonant Actuator

The Vybronics VG0832013D is a coin-type linear resonant actuator (LRA) intended for haptic feedback applications. Unlike an eccentric rotating mass motor, an LRA produces vibration

through the oscillatory motion of an internal mass near its resonant frequency, which enables faster response and more controlled haptic effects. This actuator has a diameter of 8 mm, a thickness of approximately 3.25 mm, a rated voltage of 1.8 V_{RMS}, an operating voltage range of 0.1 to 1.85 V_{RMS}, a resonant frequency of 235 Hz, a rated current of 80 mA maximum, a typical current of 58 mA, and a typical vibration force of 1.0 G under rated conditions. It is a Z-axis actuator and includes double-sided adhesive tape to facilitate mounting.

For more information check the product pages at these links: `https://www.vybronics.com/wp-content/uploads/datasheet-files/Vybronics-VG0832013D-datasheet.pdf`

5.3.1 Connections

MISSING

5.3.2 Practical notes

1. The VG0832013D should not be driven directly from a GPIO pin, since it requires an AC excitation signal rather than a simple DC drive.
2. Best performance is achieved when the actuator is driven close to its resonant frequency of 235 Hz.
3. Since this actuator includes adhesive tape for mounting, the mechanical fixation is simplified, but the mounting surface should still be flat and stable to obtain repeatable vibration performance.

5.4 Narrow Band Linear Resonant Actuator

The Vybronics VL120628H is a linear resonant actuator (LRA) designed for haptic feedback applications. Unlike a conventional eccentric rotating mass motor, an LRA produces vibration through the oscillatory motion of an internal mass at its resonant frequency, which allows faster response and more precise haptic effects. This actuator has dimensions of 12 × 6 × 2.8 mm, an X-axis vibration direction, a rated voltage of 2.0 V_{RMS}, a resonant frequency of 200 ± 10 Hz, a rated current of 130 mA_{RMS} maximum, and a typical acceleration of 1.8 ± 0.5 G under rated conditions

For more information check the product pages at these links: `https://www.vybronics.com/wp-content/uploads/datasheet-files/Vybronics-VL120628H-datasheet.pdf`

5.4.1 Connection

MISSING

5.4.2 Practical notes

1. The VL120628H should not be driven directly from a GPIO pin, since it requires an AC excitation signal rather than a direct DC drive.
2. Best performance is achieved when the actuator is driven close to its resonant frequency of 200 Hz.
3. A dedicated haptic driver is recommended for reliable operation and proper waveform generation.
4. The mounting condition affects the vibration response, so the actuator should be fixed carefully in the final mechanical structure.

5.5 Thermoelectric Peltier Module

The Same Sky CP40236 is a thermoelectric cooler (TEC), also known as a Peltier module. It generates a temperature difference between its two faces when a DC current is applied: one side becomes hot while the other becomes cold. Reversing the current direction swaps the hot and cold sides. The device is a single-stage module with dimensions of 20 × 20 × 3.6 mm, wire leads, a maximum current of 4 A, a maximum voltage of 8.6 V, an internal resistance of approximately 2 Ω, and a maximum temperature difference of about 66 °C under specified operating conditions

For more information check the product pages at these links: `https://www.sameskydevices.com/product/resource/cp40.pdf`

5.5.1 Connections

Since the CP40236 requires significantly more current than the Black Pill STM32F411 can provide directly, it must be driven through a motor driver and powered using an external supply. In this setup, an L298N motor driver board is used. The L298N is a dual H-bridge driver, so one bridge channel can be used to drive the Peltier module as a two-terminal load.

- **Peltier terminal 1 (red) → L298N OUT1**
- **Peltier terminal 2 (black) → L298N OUT2**
- **External supply positive → L298N motor supply input**
- **External supply ground → L298N power ground**
- **Pixi Click MAX11300 GPO → L298N IN1**
- **Pixi Click MAX11300 GPO → L298N IN2**

- **L298N ENA → kept enabled with onboard jumper**

The two controller signals connected to **IN1** and **IN2** define the polarity applied to the Peltier module. When the logic states are reversed, the direction of the current through the module is reversed, which swaps the hot and cold sides. In this configuration, the channel is continuously enabled through the ENA jumper, so no additional enable signal is required from the microcontroller.

5.5.2 Practical notes

1. A heat sink should be attached to the hot side of the Peltier module; otherwise, thermal performance decreases rapidly and overheating may occur.
2. For accurate thermal regulation, a temperature sensor such as a thermistor can be added and used in a closed-loop control system.
3. The Peltier module must not be powered directly from the Black Pill 3.3 V output or from USB.
4. The L298N board must be supplied from an external power source sized for the Peltier module.
5. The selected driver should be checked carefully, since Peltier modules can draw substantial current and the L298N also introduces voltage drop across the bridge.

References